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CO4 AT2 INDUSTRY PROBLEM SOLVING TASK

- 1. A cloud storage company is facing performance issues due to frequent file access delays. Analyse whether contiguous, linked, or indexed allocation would be most suitable for improving access speed and justify your choice with reasons.**

For a cloud storage company facing performance issues due to frequent file access delays, indexed allocation would generally be the most suitable choice.

Indexed allocation stores all pointers to a file's data blocks in a separate index block. Therefore, the file's data blocks do not need to be contiguous, and files can grow without requiring relocation.

Reasons:

- It supports both sequential and direct access.
- It avoids external fragmentation.
- Files can expand dynamically.
- Data blocks can be located anywhere on the disk.
- An index block makes it easier to locate the required data blocks.

Contiguous allocation provides excellent access speed, but file growth can be difficult and may cause external fragmentation. Linked allocation supports file growth but is slower for direct access because pointers must be followed sequentially.

Conclusion: Indexed allocation offers the best balance between flexibility, scalability, and access performance for cloud storage.

2. A data center experiences high disk head movement while processing user requests. Suggest the most appropriate disk scheduling algorithm (FCFS, SSTF, SCAN) to reduce seek time and explain how it improves performance in this scenario.

The most appropriate algorithm is SSTF (Shortest Seek Time First).

SSTF selects the pending disk request closest to the current position of the disk head. This reduces the distance travelled by the disk head and consequently lowers average seek time.

How it improves performance:

- Serves nearby requests first.
- Reduces unnecessary head movement.
- Decreases average seek time.
- Improves disk throughput.
- Provides faster response for most requests.

For example, if the disk head is at cylinder 50 and pending requests are at cylinders 10, 45, 55, and 90, SSTF will serve 45 or 55 first because they are closest to the current head position.

However, SSTF may cause starvation for requests located far away. If fairness and bounded waiting are more important, SCAN may be preferred.

Conclusion: SSTF is suitable when the main objective is reducing seek time and improving immediate performance.

3. **An operating system uses a bitmap for free space management but struggles with slow allocation when the disk size increases. Propose an optimization technique to improve the efficiency of bitmap-based allocation and justify your solution.**

The operating system can improve bitmap allocation by using hierarchical or word-level bitmap management with bit manipulation.

In a basic bitmap, each disk block is represented by one bit. When the disk becomes large, searching the entire bitmap for free blocks can be slow. To improve this, the system can divide the bitmap into words or groups and maintain a higher-level bitmap indicating which groups contain free blocks.

Optimization technique:

1. Divide the bitmap into fixed-size words or groups.
2. Maintain a summary bitmap for those groups.
3. Search the summary bitmap first.
4. Scan only the selected group for a free block.
5. Use processor operations such as “find first zero bit” to locate free space quickly.
6. Cache recently used free-space information.

Advantages:

- Reduces the search area.
- Speeds up free-block allocation.
- Uses less processing time.
- Works efficiently for large disks.
- Preserves the low storage overhead of bitmap allocation.

Conclusion: A hierarchical bitmap combined with word-level bit operations is an efficient solution for improving bitmap-based allocation.

4. **A real-time application requires consistent and predictable disk access times. Evaluate whether SCAN or C-SCAN would be more suitable and explain your reasoning based on the system requirements.**

For a real-time application requiring consistent and predictable disk access times, C-SCAN (Circular SCAN) is generally more suitable.

C-SCAN services requests in one direction only. When the disk head reaches one end, it returns directly to the beginning without servicing requests during the return trip. It then continues servicing requests in the same direction.

Why C-SCAN is suitable:

- Provides more uniform waiting times.
- Reduces variation in response time.
- Prevents requests near the middle from receiving repeated preference.
- Gives a more predictable service pattern.
- Is appropriate when fairness and consistent access are important.

SCAN services requests in both directions, like an elevator. It may provide good average performance, but requests at different disk locations may experience different waiting times.

Conclusion: C-SCAN is preferable when predictable and consistent access times are more important than slightly lower average seek time. For strict real-time systems, additional real-time scheduling mechanisms may also be required because neither SCAN nor C-SCAN guarantees a hard deadline.

5. A company wants to design a hybrid storage system combining file allocation and disk scheduling techniques. Suggest a suitable combination (allocation + scheduling) for handling large-scale data access efficiently and justify your choice based on performance factors.

A suitable hybrid design is:

Indexed allocation combined with SCAN disk scheduling.

Indexed allocation allows files to grow dynamically and supports direct access, while SCAN reduces excessive disk-head movement by servicing requests in both directions.

Why indexed allocation is suitable:

- Supports large files.
- Allows noncontiguous storage.
- Provides direct access to file blocks.
- Avoids external fragmentation.
- Is flexible for cloud and enterprise storage.

Why SCAN is suitable:

- Services requests according to the disk-head direction.
- Reduces unnecessary movement compared with FCFS.
- Provides better fairness than SSTF.
- Avoids indefinite starvation.
- Offers a good balance between throughput and response time.

The system can further improve performance by placing the index blocks and frequently accessed data in faster storage, using caching, and grouping requests by nearby disk locations.

Conclusion: Indexed allocation with SCAN scheduling provides a balanced solution for large-scale storage because it combines flexible file placement with efficient and fair disk access.